Concept sheet: Fooled by Randomness, Part I, Ch. 1–2

Use the red sheet over the red text. Answer in your head, then lift the sheet to check. Page references are printed book pages. Pages 1–4: questions; then vocabulary and uses; then the review key.

Part I intro: Croesus and Solon (pp. 9–10)

1. [key move]
Solon's warning: what came with the help of luck can be lost to luck, rapidly and unexpectedly; what came with little luck is more resistant to randomness.

2. [definition]
Taleb's own names, in this book, for the problem of induction: the black swan, or the rare event.

3. [in words]
The skewness issue: it does not matter how often something succeeds if failure is too costly to bear.

4. [fact]
Yogi Berra's modern version of Solon: “it ain’t over until it’s over”. The Croesus–Solon meeting itself was chronologically impossible (legend).

Chapter 1: Nero, John and the dentist (pp. 11–25)

5. [key move]
Nero's objective is not to maximize profit but to avoid having trading taken away from him (“shooting for longevity”).

6. [definition]
In trader lingo, to blow up means not just to lose money but to lose far more than ever expected, to the point of being thrown out of the business.

7. [key move]
Nero exits trades after a predetermined loss and never exposes himself to losing more than about $1 million whatever the probability of such an event.

8. [in words]
Nero's payoff shape: small, frequent losses and rare large winners; he wants to benefit from panics and crashes, not be wiped out by them.

9. [trap]
Nero holds only Treasury bonds so that his wealth depends on his cash earnings, not on market valuations. Cash-flow-negative startup owners instead depended on the randomness of the market's election of the winner.

10. [contrast]
Trading forces you to think hard; people who merely work hard end up focusing on noise rather than signal.

11. [trap]
John looked richer than Nero but carried a large hidden risk: a rare, ruinous event; he missed it because his market experience was too short and he never studied history.

12. [in words]
Taleb's image for John's business: a nap on a railway track. You earn money every month for a long time, then lose a multiple of your cumulative gains in a few hours.

13. [fact]
Preference experiment: most people prefer to make $70,000 when others make $60,000 over $80,000 when others make $90,000.

14. [predict]
Success raises serotonin, which raises apparent leadership ability and brings more success: a virtuous (positive feedback) circle, until a random kick reverses it into a vicious one.

15. [definition]
Lucky fools do not suspect they are lucky fools (by definition), and success still inflates their confidence, because our hormonal system does not know whether success came from randomness.

16. [fact]
At any time, many businessmen with outstanding track records are no better than randomly thrown darts; the least skilled can be the richest because they make no allowance for luck.

17. [key move]
Judge a profession by the average of everyone who enters it, not by the sample of those who succeeded.

18. [predict]
Replay Nero's life millions of times: few paths are marred by bad luck, but also few have extreme good luck. Replay the lottery-winning janitor's life a million times: about one path in a million wins.

19. [trap]
For most people probability concerns only the future (a past event is certain). Taleb's accounting also includes the possible outcomes that were never observed.

Chapter 2 (pp. 26–39)

20. [key move]
A decision cannot be judged by its outcome alone, but by what could have happened instead.

21. [predict]
Russian roulette for $10 million: six equally likely histories, five enrich, one is an obituary, yet only one history is observed.

22. [definition]
Taleb's term for the usually unobserved process that produces the visible results: the generator.

23. [predict]
A 25-year-old plays once a year. His chance of surviving to 50 is very slim; with thousands of players you see a handful of extremely rich survivors and a very large cemetery.

24. [contrast]
$10M from roulette vs $10M from dentistry: identical purchasing power (an accountant sees no difference), but they differ in dependence on randomness.

25. [in words]
Mathematics is a way of thinking, not just a numbers game; one need not compute alternative histories, only assess their attributes.

26. [predict]
Real life is more vicious than roulette: the bullet sits among hundreds or thousands of chambers, so after a few dozen tries one forgets it exists.

27. [trap]
Unlike roulette, in reality one cannot observe the barrel; so one can play roulette unwittingly and call it a low-risk strategy.

28. [fact]
Gamblers, investors and decision makers deny such risks because they feel what happens to others would not necessarily happen to them.

29. [contrast]
Two polar types: those who never accept randomness and those tortured by it.

30. [contrast]
Scientists on Wall Street failed only slightly less often than MBAs, because many understood the letter but not the spirit of the math.

31. [trap]
Over-correction: some people trained in quantum mechanics see only alternative histories and ignore the one that actually took place.

32. [contrast]
Jean-Patrice cared only about the generator, oblivious of results. Kenny, the “bottom line” type, judged by results and was a time bomb.

33. [key move]
Jean-Patrice taught Taleb to look for invisible risks of blowup in any portfolio, and to beware mature financial institutions.

34. [trap]
We know Alexander and Caesar because they took big risks like thousands of others and happened to win. Taleb contests the quality of their strategies, not that they won. The Iliad's heroes are heroes for their behavior, not their results.

35. [trap]
George Will told Shiller that people who listened to him would have lost money. Taleb: one wrong market call carries no undue significance, and an irrational market can become even more irrational.

36. [trap]
Beware confusing correctness with intelligibility: an idea that cannot be put in a sound-bite is not a sign of a confused mind.

37. [in words]
Much of what science proved looked like lunacy when first discovered; so what sounds smart in a meeting or the media is suspicious.

38. [key move]
A risk manager who blocks a profitable trader is blamed for lost opportunity; if a blowup occurs he is responsible. So many play politics with vague memos.

39. [definition]
Doctor analogy: a false positive means saying sick when healthy; a false negative means saying healthy when sick.

40. [fact]
Taleb's paradox: he wants investors in general to remain fools of randomness, and his greatest risk is to become successful, since his business would then disappear.

Vocabulary (standard terms) — term → meaning

Blow-up / risk of ruin	a loss so large it ends your ability to keep playing
Drawdown	peak-to-trough fall in capital
Tail risk / rare event	low-probability, high-impact outcome far from the average
Black swan	rare, high-impact surprise that past data did not warn of
Problem of induction	past observations cannot prove a rule (“no failure yet” ≠ “safe”)
Skewness	asymmetry of outcomes: many small gains vs rare huge loss (or the reverse)
Naked option selling	collecting small premiums while exposed to very large losses
Survivorship bias	judging from the winners that remain, ignoring those that vanished
Sample path / alternative history	one possible realized run of a random process
Monte Carlo simulation	replaying a random process many times to see the range of outcomes
Data-generating process (“generator”)	the mechanism that produces the observed results
Ex ante / ex post	judged before the outcome is known / after
Outcome (results) bias	judging a decision by how it turned out
Signal vs noise	meaningful information vs random fluctuation
Type I / Type II error	false positive / false negative

Symbols (none appear in pp. 7–39; common in the field) — symbol → name

E[X]	expected value: probability-weighted average of all outcomes
P(A)	probability of event A
μ	mean (average)
σ	standard deviation (typical size of fluctuation / volatility)

Where it shows up — *my additions, not in the book*

• Trading bot (mean reversion): many small wins with one rare large loss has the railway-track profile. Judge it by max drawdown and a fixed stop rule (Nero's predetermined loss), not by win rate.
• Options: selling naked options is John's profile; owning convexity (small losses, rare large wins) is Nero's.
• Backtests: a single history is one sample path. Replay many paths (Monte Carlo) before trusting a result.
• Content on X / Pixiv: one viral post is not proof of skill. Judge the process (the generator), and distrust “what worked” advice from survivors.
• Later in the book: Ch.3 Monte Carlo, Ch.6 skewness, Ch.7 induction (the book's own forward references).

Review key

Answer in red, then the reason and page.

1. [key move] can be lost to luck, rapidly and unexpectedly / more resistant to randomness
p.10. Luck-made gains are fragile; skill-made gains are sturdier.

2. [definition] the black swan, or the rare event
p.10. Standard name: black swan / tail event.

3. [in words] how often something succeeds / failure is too costly to bear
p.10. Frequency of wins is not the test; size of the worst loss is.

4. [fact] "it ain't over until it's over" / chronologically impossible (legend)
p.10. The Berra line has the advantage of being true.

5. [key move] to avoid having trading taken away from him ("shooting for longevity")
p.15. Survival first, profit second.

6. [definition] to lose far more than ever expected, to the point of being thrown out of the business
p.15. Like a doctor losing a licence or a lawyer being disbarred.

7. [key move] after a predetermined loss / whatever the probability of such an event
p.15. The cap holds regardless of probability.

8. [in words] small, frequent losses and rare large winners / benefit from panics and crashes
pp.15–16. "I love taking small losses… I just need my winners to be large."

9. [trap] his cash earnings, not on market valuations / the randomness of the market's election of the winner
p.16. He missed the bull market but has no exposure to a bear market.

10. [contrast] think hard / noise rather than signal
p.17. Hard work without thinking drowns you in randomness.

11. [trap] a rare, ruinous event / his market experience was too short and he never studied history
pp.18, 20. Not seeing a risk is different from not having it.

12. [in words] a nap on a railway track / every month for a long time / a multiple of your cumulative gains in a few hours
p.21. Same shape as option sellers wiped out in 1987, 1989, 1992, 1998.

13. [fact] $70,000 when others make $60,000 / $80,000 when others make $90,000
p.20. Relative rank (pecking order) beats absolute wealth.

14. [predict] a virtuous (positive feedback) circle / a random kick
pp.22–23. Failure then lowers rank, which causes further drops.

15. [definition] do not suspect they are lucky fools / does not know whether success came from randomness
p.22.

16. [fact] no better than randomly thrown darts / they make no allowance for luck
p.22.

17. [key move] the average of everyone who enters it / the sample of those who succeeded
p.24. This is why a dentist beats a rock musician in expectation.

18. [predict] few paths are marred by bad luck, but also few have extreme good luck / about one path in a million wins
pp.24–25. Narrow range of outcomes = resistance to randomness.

19. [trap] the future / the possible outcomes that were never observed
p.25.

20. [key move] by its outcome alone, but by what could have happened instead
p.26. Easy to say, hard to practise.

21. [predict] six equally likely histories, five enrich, one is an obituary / only one history is observed
p.27. The winner gets admired; the five other histories stay invisible.

22. [definition] the generator
p.27. Standard name: data-generating process.

23. [predict] very slim / a handful of extremely rich survivors and a very large cemetery
p.27. Bad histories catch up with repeat players.

24. [contrast] identical purchasing power (an accountant sees no difference) / dependence on randomness
p.28. Alternative accounting.

25. [in words] a way of thinking, not just a numbers game / assess their attributes
p.28. Probability is a qualitative subject.

26. [predict] hundreds or thousands of chambers / forgets it exists
p.28. False sense of security.

27. [trap] cannot observe the barrel / a low-risk strategy
pp.28–29. We see the wealth, never the process or the losers.

28. [fact] what happens to others would not necessarily happen to them
p.28. Taleb's name: denigration of history.

29. [contrast] never accept randomness / tortured by it
p.30. Flat “business orientation” people dropped like flies when payoffs became counterintuitive.

30. [contrast] only slightly less often than MBAs / understood the letter but not the spirit of the math
p.31. Lacking practical intelligence, on average.

31. [trap] only alternative histories and ignore the one that actually took place
p.30.

32. [contrast] the generator, oblivious of results / results / a time bomb
pp.32–34. Kenny's star traders were “random gunslingers”.

33. [key move] invisible risks of blowup in any portfolio / beware mature financial institutions
p.34.

34. [trap] happened to win / the quality of their strategies / for their behavior, not their results
p.35.

35. [trap] carries no undue significance / can become even more irrational
p.36. Mixing forecast with prophecy: making and unmaking prophets by the roulette wheel.

36. [trap] correctness with intelligibility / a confused mind
p.37. MBAs are trained to over-simplify.

37. [in words] looked like lunacy when first discovered / suspicious
p.38. Common sense is a collection of acquired misconceptions (the book's Einstein line).

38. [key move] for lost opportunity / he is responsible / play politics with vague memos
pp.38–39. Incentives make the job awkward.

39. [definition] saying sick when healthy / saying healthy when sick
p.39. Standard: Type I / Type II error.

40. [fact] to remain fools of randomness / to become successful
p.39.